Classic-mode stress test: every line is treated as math, so use `\text{...}` for prose, and keep lots of varied single-line formulas.

Inline-ish typography checks: punctuation, spacing, and long lines should wrap nicely without clipping.

$$e^{i\pi} + 1 = 0$$

$$\sqrt{2} \notin \mathbb{Q}$$

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}:\ n > N \Rightarrow |x_n - L| < \varepsilon$$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

$$\sum_{k=0}^{n} \binom{n}{k} x^k = (1+x)^n$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\int_0^1 x^2\, dx = \frac{1}{3}$$

$$\int_0^\infty e^{-ax} \cos(bx)\, dx = \frac{a}{a^2 + b^2} \quad (a > 0)$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad (x > 0)$$

$$\nabla \cdot (\nabla \phi) = \Delta \phi$$

$$\|x\|_2 = \sqrt{x^\top x}$$

$$\|x\|_\infty = \max_i |x_i|$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 2 & 1 & 0 \end{pmatrix}$$

$$\det(A) = 1((-1)\cdot 0 - 4 \cdot 1) - 2(0 \cdot 0 - 4 \cdot 2) + 3(0 \cdot 1 - (-1) \cdot 2)$$

$$x^\top A x = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$$

$$\lambda_{\max}(A) \geq \lambda_{\min}(A)$$

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

$$\mathbb{P}(X \leq t \mid \mathcal{F}_{\tau \wedge \sigma_{\text{very_long_stopping_time}}}) = \mathbb{E}[\mathbf{1}_{\{X \leq t\}} \mid \mathcal{F}_{\tau \wedge \sigma_{\text{very_long_stopping_time}}}]$$

$$\left(\prod_{j=1}^{m} \left(1 + \frac{1}{j}\right)\right) = m + 1$$

$$\sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} \to \ln 2$$

$$\exp\left(-\int_0^1 \frac{t^2}{1+t}\, dt\right) \in (0,1)$$

Long line torture test (wrap + spacing + nested delimiters): $\left(\frac{\sum_{k=1}^{n}\left(\frac{k^2+1}{k^2+k+1}\right)}{\prod_{j=1}^{m}\left(1+\frac{1}{j}\right)}\right) \cdot \left(\int_0^\infty \frac{\cos(bx)}{1+x^2}\, dx\right) \leq \left(\frac{n}{m+1}\right) \cdot \left(\frac{\pi}{2} e^{-|b|}\right)$

More symbol variety: $\angle ABC = \theta,\ \alpha + \beta + \gamma = \pi,\ \exists!\, x \in \mathbb{R}:\ x^2 = 2,\ \partial_t u = c^2 \Delta u$

End of classic-mode file.